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UNITS :-

- ① Engineering Materials
 - glass
 - Cement
 - Lubricants
- ② Corrosion
- ③ Water
- ④ Organic Fuel
- ⑤ Green chemistry

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Engineering Materials.

Glass

→ Introduction of glass → Glass is hard, amorphous, transparent and highly viscous liquid. It has low sharp melting point. It is a good electrical insulator, it can absorb, reflect and transmit light.

* → Vitrification → Glass is a transparent amorphous solid. When glass is produced by heating the ingredients, a highly viscous transparent liquid is produced. The liquid on slow cooling produce an amorphous solid and non crystalline. This process is known as Vitrification.

* → De-Vitrification → When the glass is soften by heating and is kept as such for a long time, it appear to loose its transparency due to crystallization of individual silicates. This property of glass is known as De-Vitrification.

Manufacturing of Glass:-

Constituents

Effect

Na_2O

Melting of silica and reduce viscosity.

CaO

Makes glass resistant to water & chemical

K_2O

Increase softening temp of glass.

PbO

Increase refractive index of glass

Fe_2O_3

Do not transmit U.V light.

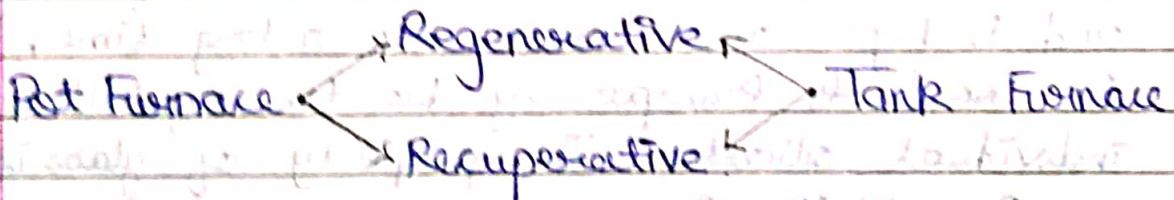
→ Raw material →

- (i) Sodium as Na_2CO_3
- (ii) Potassium as K_2CO_3
- (iii) Calcium as Limestone
- (iv) Lead as litharge
- (v) Silica as quartz
- (vi) Cullet (Broken pieces of glass)
- (vii) Colouring agents (Fertic salt - Yellow colour
Nickel - Red
Magnesoxide - Purple)

→ There are four steps involve in this process.

- ① melting ② forming & shaping ③ Annealing ④ Finishing.

① Melting:- It can be carried out



(i) Regenerative → Utilization of heat of waste gases for heating the upcoming fuel and air. The direction of waste gas & fuel are changed.

(ii) Recuperative → Heat of waste gases is utilize in heating the furnace but the incoming gas flow continuous in one direction.

(iii) Pot Furnace → At a one time many type of glass is produced. in

(iv) Tank Furnace $\Rightarrow$ At a one time only one type of glass is produced in large amount.

The Raw material are mixed in proper proportion & mixed with cullets. This are fine powder and mixture is called Batch and fuse in tank furnace. Heating is done by producer gas and air ~~mixture~~ mixture over the charge.

(A) For soft glass $\Rightarrow$

Batch $\Rightarrow$ Na_2CO_3 , CaCO_3 , SiO_2

At $< 600^\circ\text{C}$ moisture & oxide of Pb, B, As & chloride of Na, Ca, K are evaporated.

- At $700 - 900^\circ\text{C} \Rightarrow \text{Na}_2\text{CO}_3 + \text{SiO}_2 \rightarrow \text{Na}_2\text{SiO}_3 + \text{CO}_2$
- At $900 - 960^\circ\text{C} \Rightarrow \text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- At $1000^\circ\text{C} \Rightarrow \text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

(B) For Flint glass $\Rightarrow$

Batch $\Rightarrow$ PbO_2 , K_2CO_3 , SiO_2

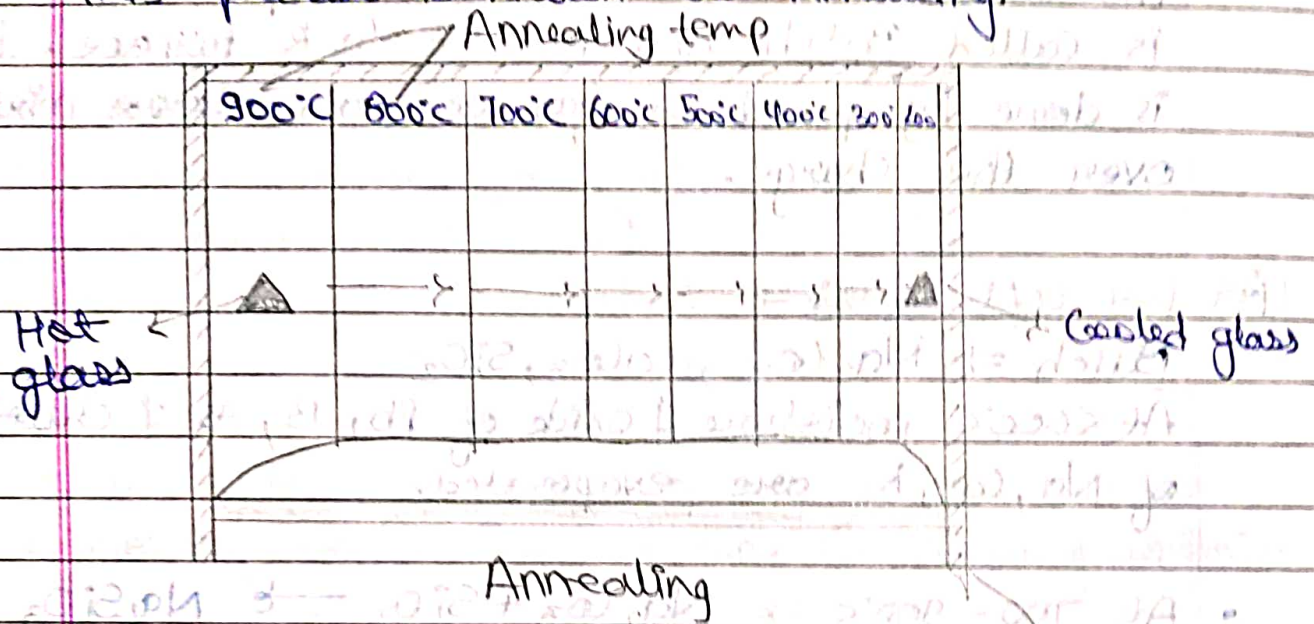
- At $600^\circ\text{C} \Rightarrow \text{Pb}_3\text{O}_4 \rightarrow 3\text{PbO} + \text{O}_2$
- At $650 - 730^\circ\text{C} \Rightarrow \text{PbO} + \text{SiO}_2 \rightarrow \text{PbSiO}_3$
- At $750^\circ\text{C} \Rightarrow \text{K}_2\text{CO}_3 + \text{SiO}_2 \rightarrow \text{K}_2\text{SiO}_3 + \text{CO}_2$
- At $850^\circ\text{C} \Rightarrow \text{PbSiO}_3$ dissolve in K_2CO_3

(2) Forming & Shaping $\Rightarrow$ Molten glass is shaped into desired shape.

(3) Annealing $\Rightarrow$ Glass is a bad conductor of heat so rapid cooling, makes the outer surface to cool fast.

Annealing is a process used in glass making to relieve internal stress, cracks. By doing Annealing process on glass the glass will not break as long as it is first.

then the internal portion. This unequal cooling causes cracks. So the glass obtained through the melting is changed into desired shape and cool very slowly to reduce the non-uniformity and cracks. This process is known as Annealing.

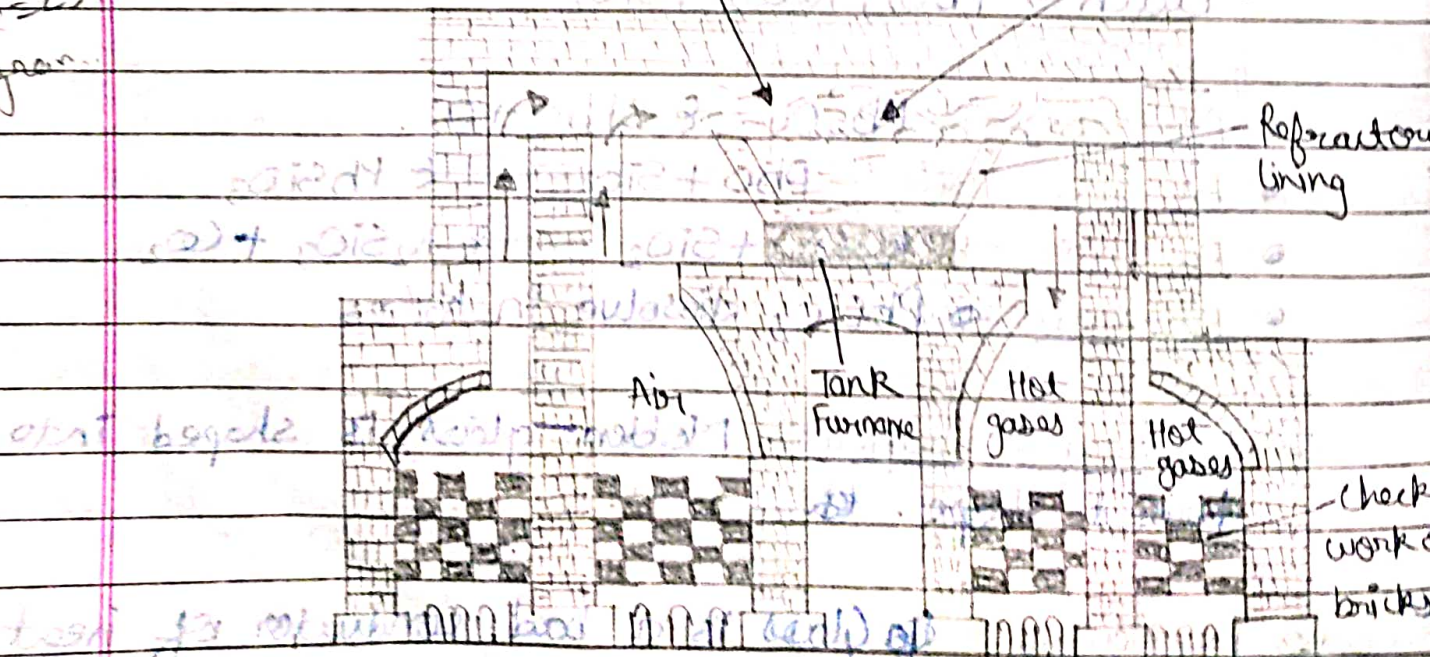


④ Finishing After Annealing all types of glass goes to finishing process like cutting, polishing, colouring and shaping.



Soda ash + limestone + sand + cullets.

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Diagram



Manufacturing of glass.

* --> Types of glass :- (i) Soda lime glass - (soft glass).

(i) Soda lime glass (soft glass) => The raw materials are silica, sodium carbonate & calcium carbonate. approximate composition is $\text{Na}_2\text{O} \cdot \text{CaO} \cdot 6\text{SiO}_2$.

--> Properties => (i) High softening point (temp)
(ii) High chemical resistant

--> Application => (i) Kitchen ware (ii) Electric bulb
(iii) bottles.

(ii) Hard glass => This type of glass is also known as potash-glass. This is obtained from silica, calcium carbonate & potassium carbonate. The approximate composition is $\text{K}_2\text{O} \cdot \text{CaO} \cdot 6\text{SiO}_2$.

--> Properties => This type of glass is more costly and fused with difficulties.

--> Application => These glasses are used for chemical apparatus, combustion tube.

(iii) Borosilicate Glass => It is most common of the hard-glass. Such glass contain silica and boron with small amount of alumina and some alkali oxides. The approximate composition is

Component	%
SiO_2	80
B_2O_3	13.5
Al_2O_3	3

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K_2O	3
Na_2O	0-5

- Properties

① Borosilicate glass have softening point.

- Application

① This type of glass mainly used in laboratory.

② They are used in industry's for pipe-line corrosive liquid.

=> Lubricant

Introduction = Any substance introduced b/w two moving / sliding surface to reduce the friction is known as lubricant.

The process reducing the friction b/w two moving surface / slide by the introduction of lubricant is known as lubrication.

• Function of lubricants

- (i) Lubricant prevent metal to metal contact, reducing the friction and the surface deformation.
- (ii) It prevents the loss of energy due to movement of metallic surface.
- (iii) It is act as coolant.
- (iv) It increase the efficiency of machine.
- (v) It reduce the running and maintenance cost.

• Classification of lubricant

Lubricants

Liquid

- Vegetable / Animal
- Mineral oil
- Synthetic oil
- Blended oil

Solid

- Graphite
- Molybdenum
- disulphide

Gemisolid

- Grease

Emulsion

- oil in water
- water in oil

① Liquid lubricants

① Animal / Vegetable oil ⇒ These were the only sources of lubricants when lubricants originate from mineral sources were not available. The most important quality of these oil is oiliness. It help in sticking to the machine parts but these oil undergoes oxidation and hydrolysis easily. For ex^o olive oil, palm oil, castor oil.

② Mineral oil ⇒ They are also known as petroleum oil since they are obtained from distillation process. This oil have different properties depending upon their boiling point. The number of carbon atom also ranges from 12-50, Short the carbon chain length oil will have lower viscosity. For ex^o light oil, medium oil, heavy oil.

③ Synthetic oil ⇒ The oil like vegetable and mineral are used at extreme condition like high temperature, pressure and high speed. The oil will decompose and make oil fume. So special lubricants are used in ~~hot~~ ^{hot} rolling mills, in furnace equipment etc. Synthetic lubricants are help at extreme condition and produce by organic and inorganic chemicals. For ex^o silicon phosphate and esters.

Synthetic oil have inflammability, high flash point, high viscosity index but some of them are toxic and corrosive in nature.

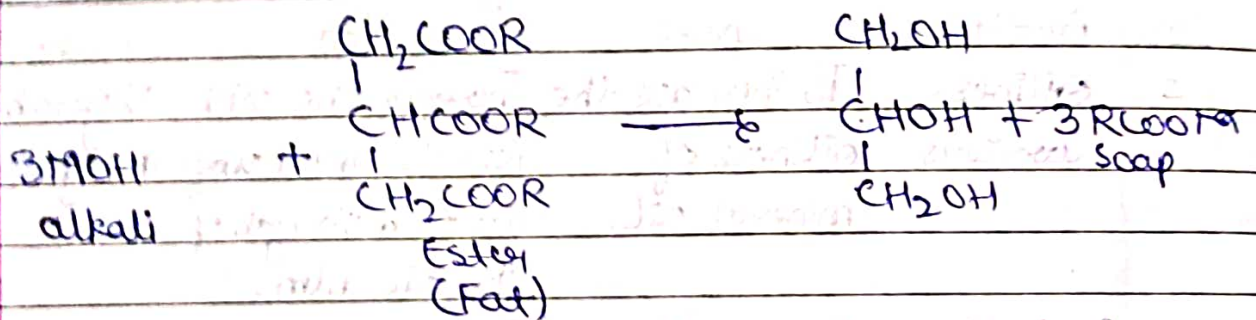
(9v) Blended oil = Animal, vegetable and mineral oil cannot be satisfactory lubricant in many of the machineries. If the mineral oil is mixed with some additives, the properties can be improved so that it can be a better lubricant.

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Table : Lubricant Additives and their functions:

Sn.	Additive	Purpose	Function	Examples
1.	Oiliness Carriers	To Increase the oiliness of mineral oil.	Increase the stickiness to metal surface and strength of the oil film.	Vegetable oils and fatty acids.
2	Antioxidants	Retard oxidative decomposition.	Terminate free radical chain reactions & decompose peroxides.	Aromatic amines, Phenol etc.
3.	Corrosion inhibitor	Prevent rusting and corrosion of metal parts in contact with the lubricant.	Neutralization of corrosive acid and provide a protective film.	Metal Sulfonates fatty acids and amines.
4	Anti foaming	Prevent formation of stable foam	Speed collapse of foam by reducing surface tension.	Silicon Polymer.
5	Viscosity Index Improver	Improve the Viscosity index of lubricant	Polymer expand with increasing temperature	High molecular weight compound like hexanol.
6	Pour point depressant	Enable lubricant to flow at low temperatures	Prevent the separation of wax from the oil.	Phenol, chlorinated wax with naphthalene.
7	Emulsifiers	Help in the formation of emulsions of lubricating oils with water	Works as a thickening agent	Sodium salt of Sulphonic acid, fatty amine salt.
8	Thickener	Increase the viscosity	It give the lubricating oil of a higher viscosity	Polystyrene, Polyester.
9	Extreme Pressure additives	For the lubricant works under the high pressure & high speed	Absorbed on metal surface or chemically react with metal	Fatty acid with sulphurized oil organic chlorine and phosphorus compound.

- (ii) Semi-solid lubricant = ~~The oil like vegetable~~
 It is obtained by dispersing soap in liquid lubricant like petroleum oil or synthetic oil.
 Additive for specific requirement are added. It is manufactured by saponification of fat by addition of lubrication oil with agitation.



The Grease can support at heavier load at lower speed. It is used for textiles and paper materials. It can act as a seal against the entry of dust, moisture but grease on storage may separate back into oil and soap. It is used in bearing, working at lower temperature. Depending upon the soap mixed it may be classified as sodium soap grease, lithium soap grease, ~~Al~~ Aluminium soap grease etc. . .

- (iii) Solid lubricant = They are preferred at different places it is used at ~~very~~ extreme operating condition where lubricant film may decompose or catch fire. Solid lubricant are used where load and temp is so high that even grease cannot work. For ex^g Graphite and Molybdenum disulphide,

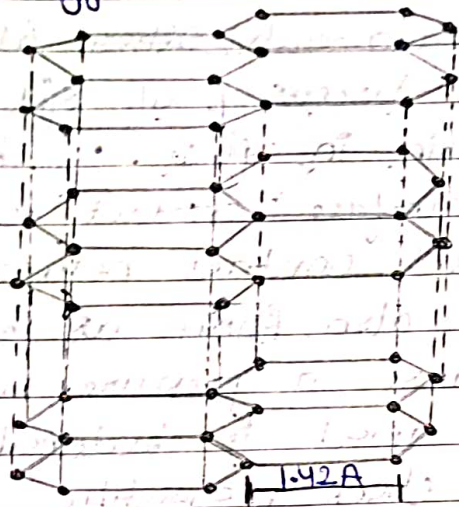
solid lubricant are ~~anisotropic~~ ^{anisotropic} in nature

- (i) Graphite consist of flat layered structure held together by weak bonds. these parallel layers slide over one-another so it act as a good lubricant.

* Imp Point

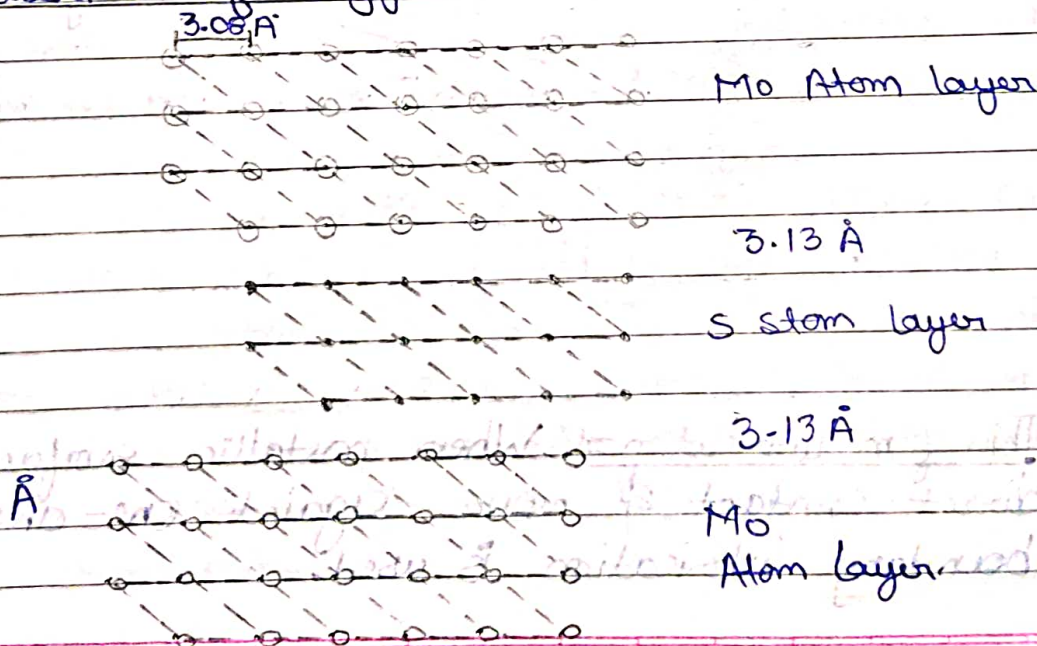
Graphite is a good lubricant in moist air, lower in oxygen and CO_2 and nil in nitrogen.

Diagram of Graphite



- (ii) Molybdenum disulphate has a sandwich like structure
* * Molybdenum disulphide is good lubricant in presence of Nitrogen & CO_2 , lower in moist air and absence of oxygen

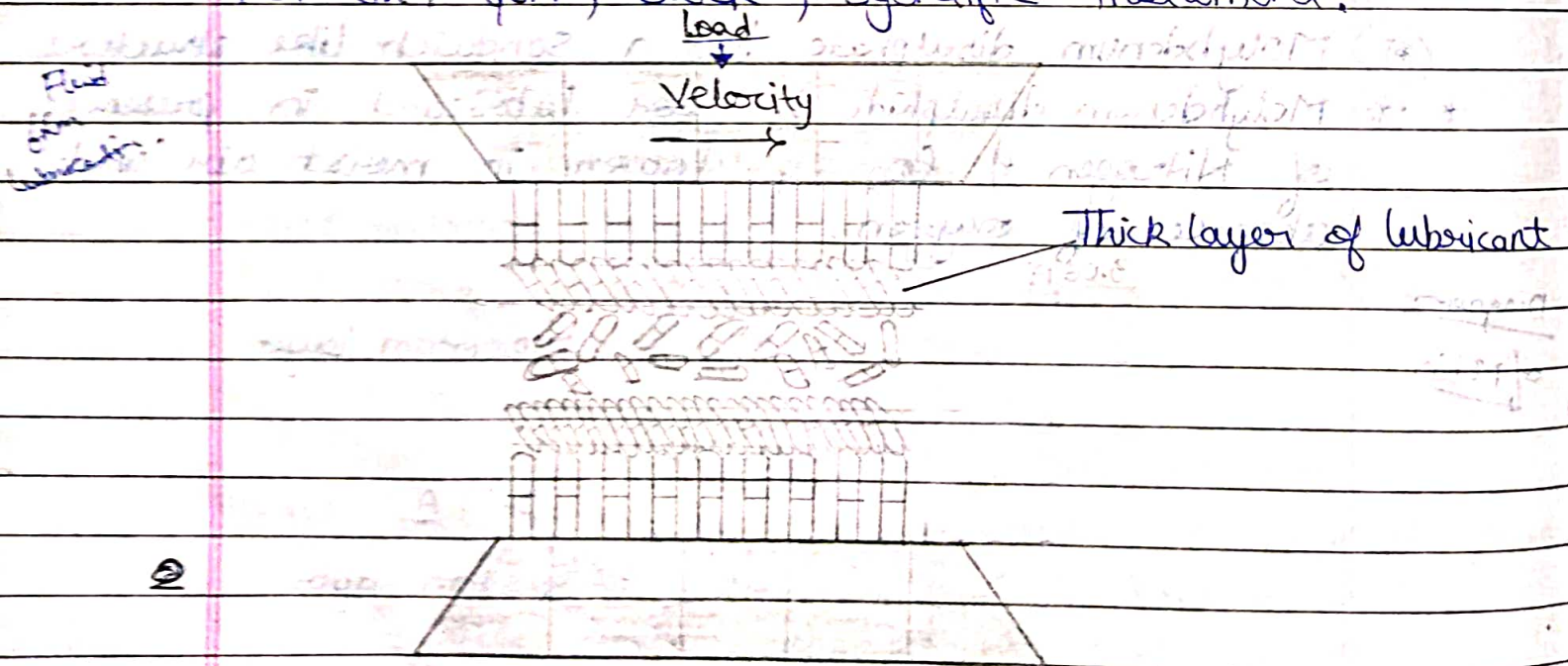
Diagram of MDS



→ Mechanism of lubrication = 6

- ① Thick layer / Fluid lubrication / hydrodynamic
- ② Thin layer / Boundary lubrication
- ③ Extreme Pressure lubrication.

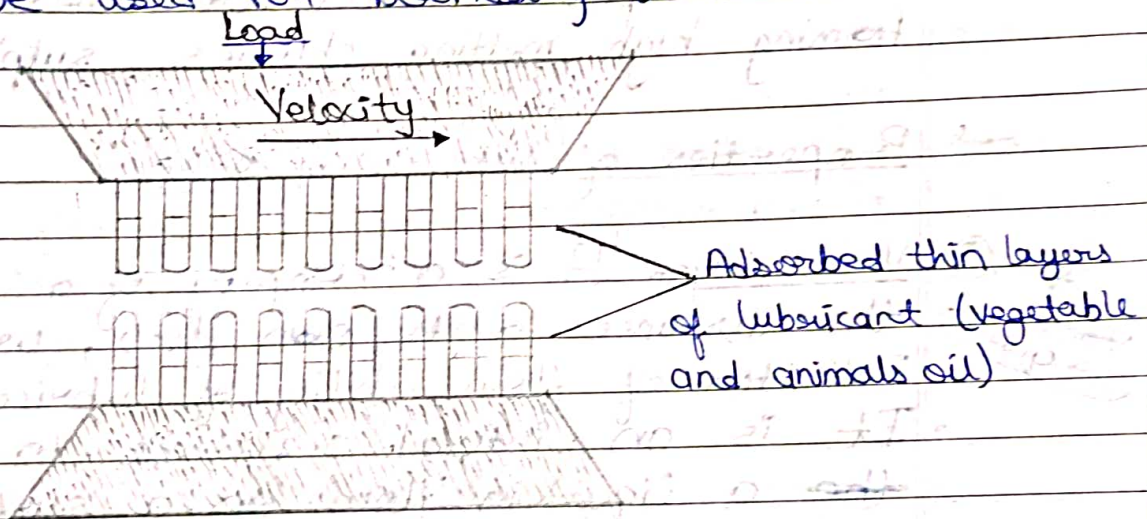
- ① Thick layer / Fluid lubrication / hydrodynamic ⇒ This type of lubrication occurs when two moving slides / surfaces are separated from each other at distance $1000 A^{\circ}$ (Aston = A°). The Viscosity of oil should be low and should remain in place. • Thick film lubricants and surface - surface contact is avoided.
- There is no direct contact between the metallic surface. It is also known as hydro-dynamic lubrication because a pressure sufficient to keep the moving surface apart is developed under such condition.
 - For cast gun, clock, scientific instrument.



- ② Thin film lubrication ⇒ When metallic surface are in direct contact & move against one-another boundary lubrication is used.

- Under such condition, a lubricant which can interact with solid surface of the metal, is used. This is
- This is the lubricant may be chemically react with the metal surface to form a permanent layer
- Vegetable & Animal oil is used for this purpose
- The solid lubricant like graphite & molybdenum disulphide either alone or mixed with oil can also be used for boundary lubrication.

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- High viscosity index, good oiliness, low pour point are some of the properties which a boundary lubricant should have.

- ③ Extreme pressure lubrication = When extreme condition like high temp & high pressure and speed is slow then the liquid lubricant may decomposed.
- The layers of lubricant are broken and the metal surfaces come into contact. Due to high speed & temp high temp the metal may be go in deformation.

- The extreme pressure condition, certain additives are added to lubricant which can form more durable film.

- Extreme pressure additives are generally

- (i) oxygen containing compound like fatty acids & esters.

- (ii) chlorine containing compounds

- (iii) sulphur containing compounds.

- These compounds react with metallic surface, forming high melting chlorides, sulphides.

→ Properties of lubricating oil =

* (i) Viscosity = It is a property of a fluid it offers the distance to its own flow, which is an indicator of flow ability of lubricating oil.

- It is an internal resistance to motion of fluid, ~~the~~ a liquid in flow on a surface consist of series of parallel layers of moving one over the another, all layers moves different velocities.

- Consider any two layer of liquid, which are at a distance of ~~small~~ 'd' moving with velocities difference 'v' then force is per unit ~~area~~ area 'F'.

- $F = \frac{\eta v}{d}$, where η = coefficient of velocity viscosity.

If $v = 1$ unit $d = 1$ unity

$$\boxed{F = \eta}$$

I = Lite
H = Heavy

Viscosity of lub then

* With decrease

The viscosity

If increase index

If high have

→ De of Ste

= The

(i) Po

(ii) N

Fe Vi S

- Viscosity determine the operating ~~so~~ characteristic of lubricant if the lubricating oil is like oil then it is easily flow.

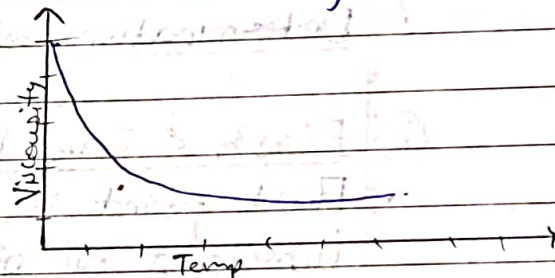
* With increase in temperature, viscosity of oil decreases. (Temp ↑ Viscosity ↓)

- The rate at which the viscosity of an oil ~~so~~ change with temperature is measured by viscosity index.

• If the viscosity of an oil falls rapidly with increase in temperature, it have low viscosity index.

• If the viscosity of an oil is only slightly effected ~~so~~ big change (↓, ↑) in temperature it has high viscosity index. A better lubricating oil will have high viscosity index.

• Viscosity $\propto \frac{1}{\text{temperature}}$



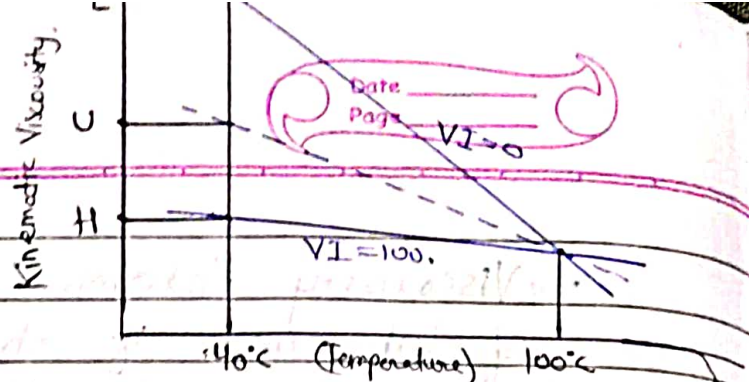
→ Determination the Viscosity index (V.I) :- For determination of Viscosity index there are two type of standard oil is used.

⇒ The variation of a liquid with temperature is called viscosity index.

- I=Light
H=Heavy
- (i) Paraffinic base Pennsylvanian oil (H-oil) of V.I = 100
 - (ii) Naphthalene base Gulf oil (L-oil) of V.I = 0.

From the list of Heavy oil, the oil which has same viscosity at ~~200~~ 210°F as the oil under test is selected. Let the viscosity of this H-oil at 100°F

is 'H'.



- From the list of L-oil, the oil which has same viscosity at ~~20~~ 210°F is selected let the viscosity of this oil at 100°F is 'L'.

$$V.I = \frac{L - H}{L - H} \times 100 \quad \text{molecular weight} \propto V.I.$$

[The viscosity of oil under test at 100°C is 'u']

- * The molecules which has linear structure exhibit high Viscosity Index. The molecules having high-molecular weight also has high V.I.
- & Determination of viscosity

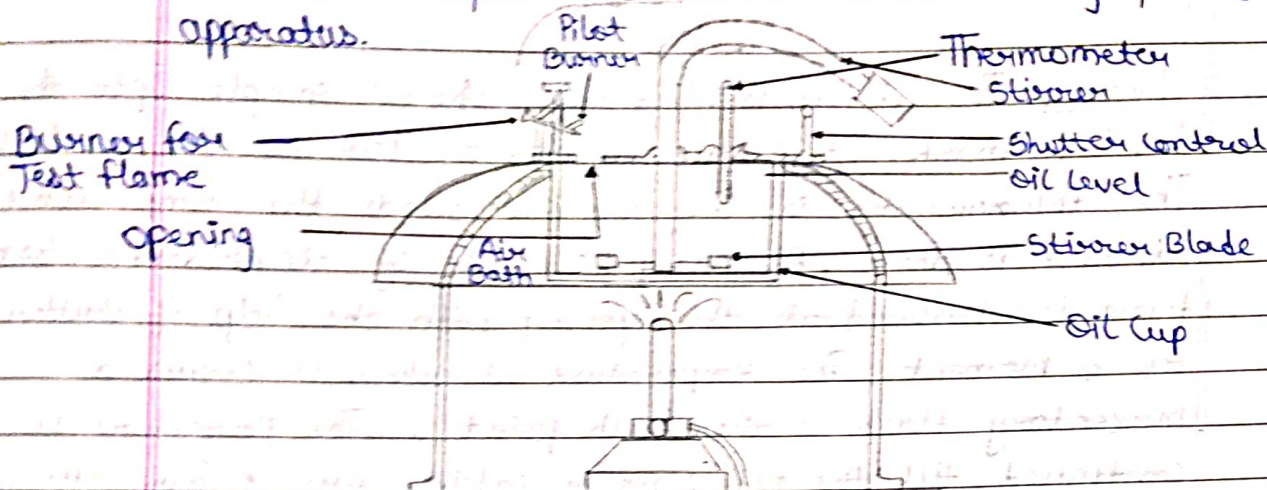
(i) Flash & Fire Point —&

- Flash point is the lowest temperature at which the lubricant oil gives enough vapours so that it is ignited for a moment.
- Fire point is the lowest temperature at which the lubricant oil gives enough vapours so that it burns continuously atleast for 5 seconds, when test flame is brought near it.

Flash point and fire point of lubricant is helpful in knowing its operating temperature. Flash and fire point should always be higher than operating temperature of the machine. It safeguards against risk of fire. Fire point in most of the cases is higher than flash point.

by $5-40^{\circ}\text{C}$.

Flash & fire point can be determined using penskymarten's apparatus.



Penskey - Marten's Flash point apparatus

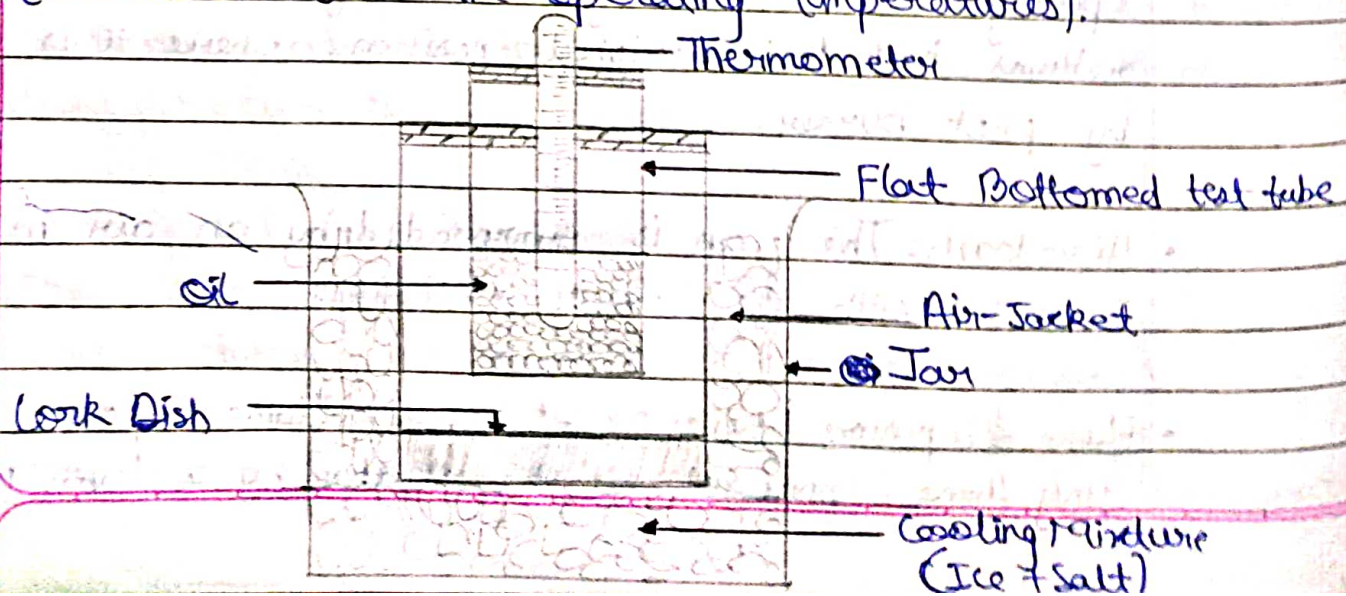
- Oil Cup: It consists of a cup made of brass (5 cm deep & 5.5 cm in diameter). The lid of the cup has 4 openings. One is for stirrer, second is for passage of air; third for thermometer & fourth for introducing test flame.
- Stirrer: The lid is provided with a stirrer having two blades.
- Shutter: The shutter is provided at the top of the cup. By moving the shutter the opening in the lid opens and flame from the flame exposure device is dipped into this opening. After introducing the test flame, the test flame returns back to its original position where it is lighted by pilot burner.
- Air Bath: The cup is surrounded by an air bath which is heated with the help of electricity.
- Flame Exposure Device & pilot burner: Flame exposure device is a tiny flame, connected to the shutter by a lever mechanism.

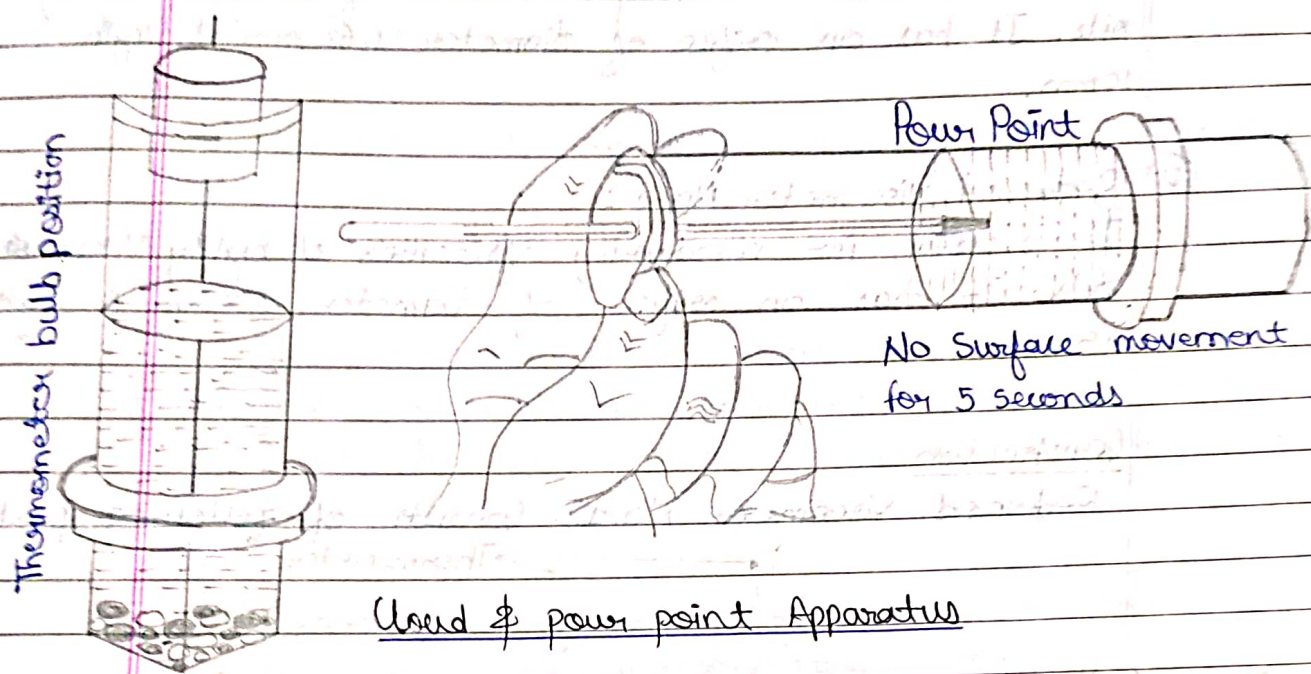
The test flame is lighted automatically with the help of pilot burner.

Working = The oil cup is filled with the oil sample upto the leveling mark. It is then heated by burning the air-bath. The thermometer in the sample reads the temperature. Stirring is done. At every 1°C rise of temperature, test flame is introduced at the opening with the help of shutter for a moment. The temperature at which it shows a momentary flash is the flash point. The procedure is continued till the oil vapour catches fire & burns for five seconds. This temperature is fire point.

(ii) Cloud & Pour Points

- When oil is cooled slowly, the temperature at which it becomes cloudy or hazy appearance, is called cloud point. While the temperature at which the oil ceases to flow is known as pour point.
- The cloud and pour points indicate the suitability of lubricants in cold conditions. The lubricant should have low cloud and pour point at the operating conditions (i.e. lower than the operating temperatures).





Cloud and pour point is determined with the help of cloud and pour point apparatus. It has a flat bottomed tube of 3cm diameter and 12cm height. This is kept in an air jacket which is cooled with the help of a freezing mixture (eg. ice + CaCl_2). The tube is half filled with oil. With the help of thermometer temperature is noted. With every degree fall of temperatures, tube is withdrawn for a moment, to find out for the cloudiness. This cooling is continued and same procedure is repeated. The temperature is noted when the oil does not flow for 5 seconds even when kept horizontal. This temperature is pour point.

(iii) Determination of Viscosity

Redwood viscometer is used to measure viscosity of a lubricating oil.

Redwood viscometer is of two types.

(1) Redwood Viscometer No.1:

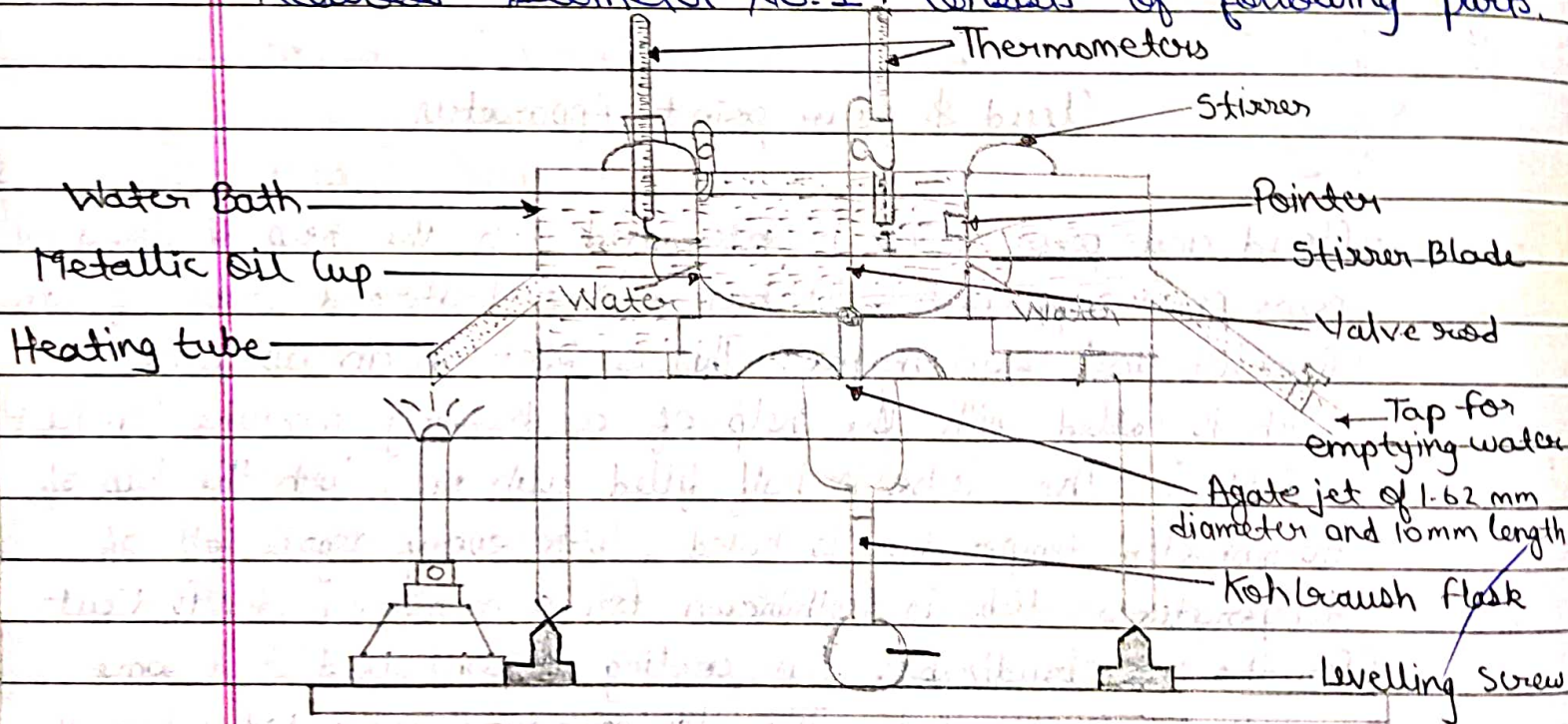
It is used for determining viscosities of thin lubricating oils. It has an orifice of diameter 1.62 mm & length 10mm.

② Redwood Viscometer No. 2:

It is used for measuring viscosities of highly viscous oils. It has an orifice of diameter 3.8 mm & length 15 mm.

Construction

Redwood viscometer No. 1 consists of following parts.



Redwood Viscometer

- oil cup: It is a silver-plated brass cylinder of 90mm height and 46.5 mm diameter. The bottom of the cylinder is fitted with an agate jet (of diameter 1.62 mm & length 10mm). The jet is opened or closed with the help of valve rod. It has a small silver-plated brass ball at the end. Oil is to be filled in the cylinder upto the

level of pointer. A thermometer is fitted to the lid which gives the oil temperature.

- Heating Bath: Oil cup is surrounded by a water bath fitted with a thermometer and a stirrer. Thermometer indicates the temperature of water.
- Stirrer: Stirrer has four blades for stirring so as to maintain uniform desired temperature. It also has a shield at the top to prevent water splashing into the oil in the cylinder.
- Kohlaraukh Flask: Below the pour hole, a vessel of 50 ml capacity is placed to collect the oil.
- Spirit level & levelling screw: For levelling of instrument, the lid of cup is provided with a spirit level. The entire apparatus rest on 3 legs provided with levelling screws.

• Working: The oil to be tested in the oil cup to the pointer level. Kohlaraukh's flask is kept below the jet outlet to collect the oil. The water bath is heated and by stirring the uniformity of temperature is maintained. Temperature of oil is noted with the help of thermometer. At desired temperature the valve from jet hole is removed. The time taken to collect 50 ml of oil in the flask is noted. The result is expressed in "Reduced No. 2 seconds" at the specific temperature. Higher the time taken for the oil to flow, higher is its viscosity.

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⇒ Cement

⇒ Introduction:-

Cement is a fine grain powder which has binding property on mixing with water. It forms a paste which can bind building blocks, stones & bricks. There are four types of cement:-

- (i) Natural Cement
- (ii) Pozzolana Cement
- (iii) Slag Cement
- (iv) Portland Cement.

(i) Portland Cement ⇒ It is also known as magic powder. It consists lime, Alumina, silica & iron. It forms paste when mixed with water. It is obtained by calcining process on mixture clay & lime containing saw material at 1500°C temperature after calcination ~~retarder~~ like gypsum is added. Portland cement was named because paste of cement with water on setting & hardening ^{are same} ~~resemble~~ in colour & hardness to portland stone ~~which~~ which is found portland island.

→ Composition of Cement ⇒ The raw material used for manufacture of Portland Cement are.

- (i) Calcareous material (eg. lime-stone, chalk)
- (ii) Argillaceous material containing silica & iron oxide & alumina (eg. clay, sand, furnace)
- (iii) Powdered coal or fuel
- (iv) Gypsum.

Constituent	Abbreviation	%	Function
Lime (CaO)	C	60-69%	Its lesser or excess amt reduces the strength of cement.
Silica	S	17-25%	Its impart strength
Alumina	A	3-8%	It makes cement quick setting it ^{weakens} excess weakens the cement.
Silica Iron oxide	F	2-4%	It provides colour, strength & Hardness
Other like alkali oxide MgO , SO_3	—	0.5-5%	SO_3 important Soundness Volume initial = Volume final

* $\rightarrow$ Manufacturing of Portland Cement $\rightarrow$ Rotary Kiln process is used for manufacturing of cement in large cement plants.

There are four steps:-

- (i) Mixing
- (ii) burning
- (iii) Grinding
- (iv) Packaging.

(i) Mixing $\rightarrow$ There are two types of process is used for mixing of raw material of cement:- (a) Dry Process (b) Wet Process.

(a) Dry Process $\rightarrow$ The Argillaceous material are crushed & powdered separately. The powder material are mixed in proper portion to get ~~separately~~ dry raw mixture. It is kept ready to store in storage bins (silos) then goes to burning process.

(b) Wet Process $\rightarrow$ The Argillaceous material are ~~for~~ crushed, powdered & stored in silos. The argillaceous material is mixed with water. The powdered lime stone & clay are goes to grinding mills.

They are mixed to form a paste is called slurry containing 40% water. Its composition is adjusted & ~~send~~ sent for burning process.

(ii) Burning zone $\Rightarrow$ For burning process rotary kiln is used. It is a ~~very~~ long steel cylinder about 2-4 meter diameter, 60-90m in length. The kiln 5-6 Degree inclined toward the exit & is rotated at a one rotation per minute. The material introduced at the upper part & passes slowly to the lower end. It passes from following different zone. ① ~~Drying~~ ^{Drying} zone ② Calcining zone ③ burning zone.

(a) Drying zone $\Rightarrow$ It is the upper part of rotary kiln where the temperature is around $100-400^{\circ}\text{C}$, water in the slurry gets evaporated.

(b) Calcining zone $\Rightarrow$ It is the central part of kiln the lime stone of dry mixture or slurry undergoes decomposition at a temp of 1000°C .

(c) Burning zone $\Rightarrow$ It is the hottest zone, temp is above 1000°C . The mixture melt & form clinkers (small, round grains). The reaction taking place are $\rightarrow$ $\text{CaO}_3 \longrightarrow \text{CaO} + \text{CO}_2$

- $2\text{CaO} + \text{SiO}_2 \longrightarrow 2\text{CaO} \cdot \text{SiO}_2$ (Dicalcium silicate) $[\text{C}_2\text{S}]$
- $3\text{CaO} + \text{SiO}_2 \longrightarrow 3\text{CaO} \cdot \text{SiO}_2$ (Tricalcium silicate) $[\text{C}_3\text{S}]$
- $3\text{CaO} + \text{Al}_2\text{O}_3 \longrightarrow 3\text{CaO} \cdot \text{Al}_2\text{O}_3$ (Tricalcium aluminate) $[\text{C}_3\text{A}]$
- $4\text{CaO} + \text{Fe}_2\text{O}_3 + \text{Al}_2\text{O}_3 \longrightarrow 4\text{CaO} \cdot \text{Fe}_2\text{O}_3 \cdot \text{Al}_2\text{O}_3$
(Tetra calcium Aluminoferrite) $[\text{C}_4\text{AF}]$

$\rightarrow$ The chemical changes taking place at different temp in manufacturing of cement $\rightarrow$

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* Temperature	Chemical changes.
$>100^{\circ}\text{C}$	Evaporation of water.
$>500^{\circ}\text{C}$	Removal of water from clay.
$>800^{\circ}\text{C}$	Decomposition of CaCO_3 & MgCO_3
$900-1100^{\circ}\text{C}$	Decomposition of CaCO_3 is complete.
$1200-1600^{\circ}\text{C}$	Formation of C_3A , C_3S , C_4AF Completed

→ The characteristics of main constituent are:→

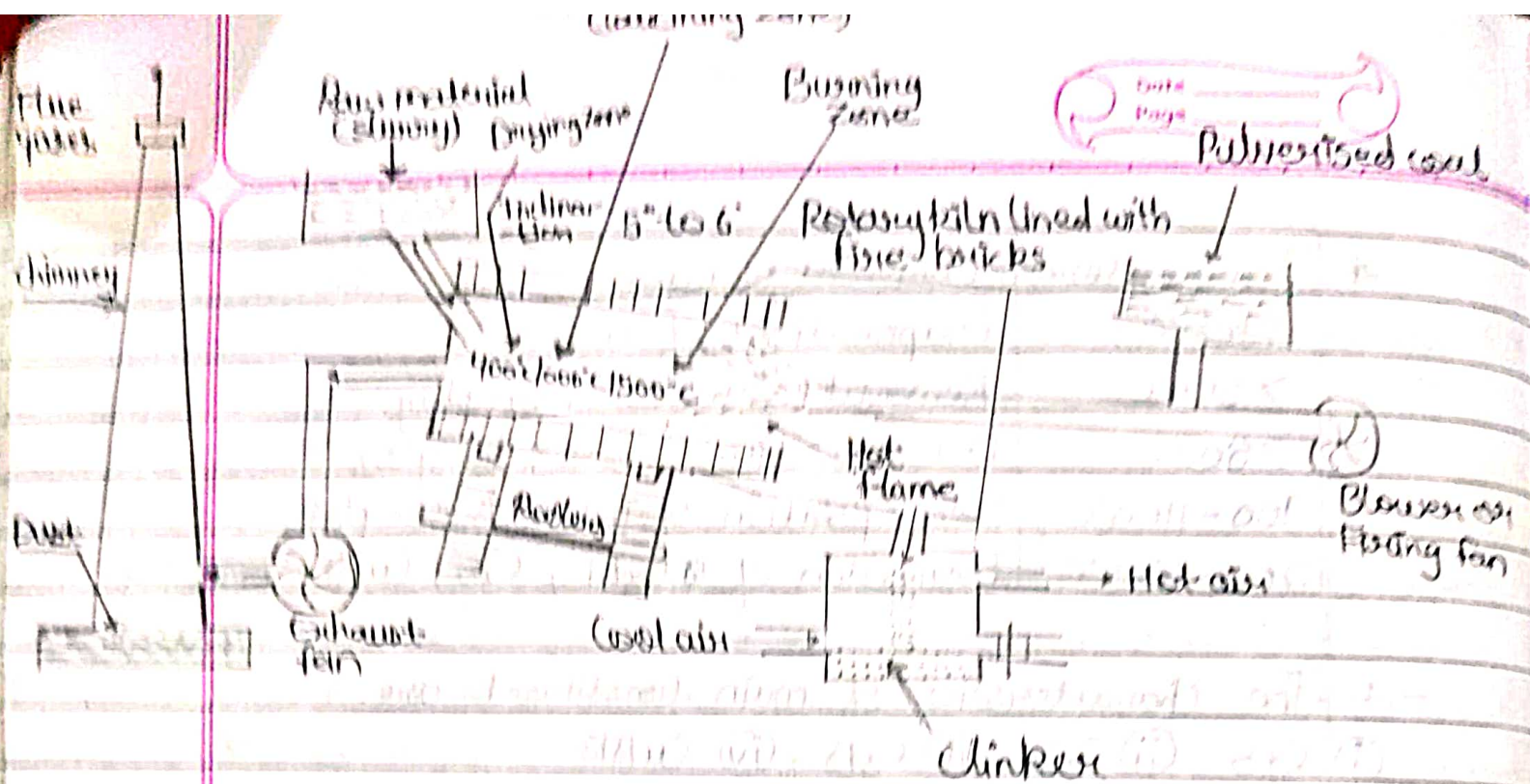
(i) C_2S (ii) C_3S (iii) C_3A (iv) C_4AF

(i) Dicalcium silicate (C_2S) ⇒ Its ultimate strength is less in compare to C_3S . It is also responsible for strength of cement.

(ii) Tricalcium silicate (C_3S) ⇒ It is responsible for ultimate strength of the cement. Its rate of hydration is medium. It develop its ultimate strength very fast.

(iii) Tricalcium aluminate (C_3A) ⇒ It undergoes hydration very fast rate. The initial set of cement is due to C_3A . Its early strength is good but ultimate strength is poor.

(iv) Tetra calcium Aluminate (C_4AF) ⇒ It has slow rate of hydration its both early & ultimate strength are poor so do not contribute much to the strength of cement.

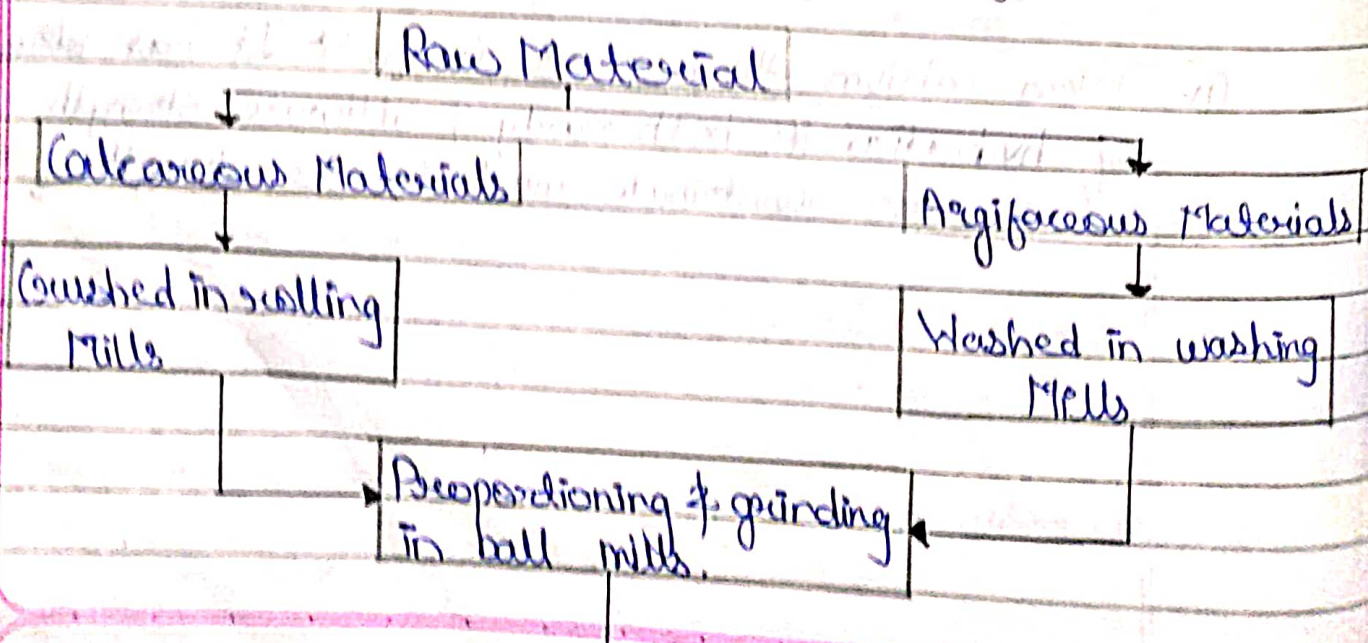


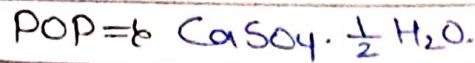
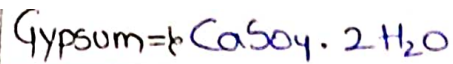
Rotary Kiln for Cement manufacture

(iii) Grinding = We get clinkers from rotary kiln. These clinkers after cooling is grinded & mixed with 2-3% gypsum. Gypsum increases the initial setting period of cement.

(iv) Packing = The brown fine powder of cement is stored in silos & then packed into 15kg bags.

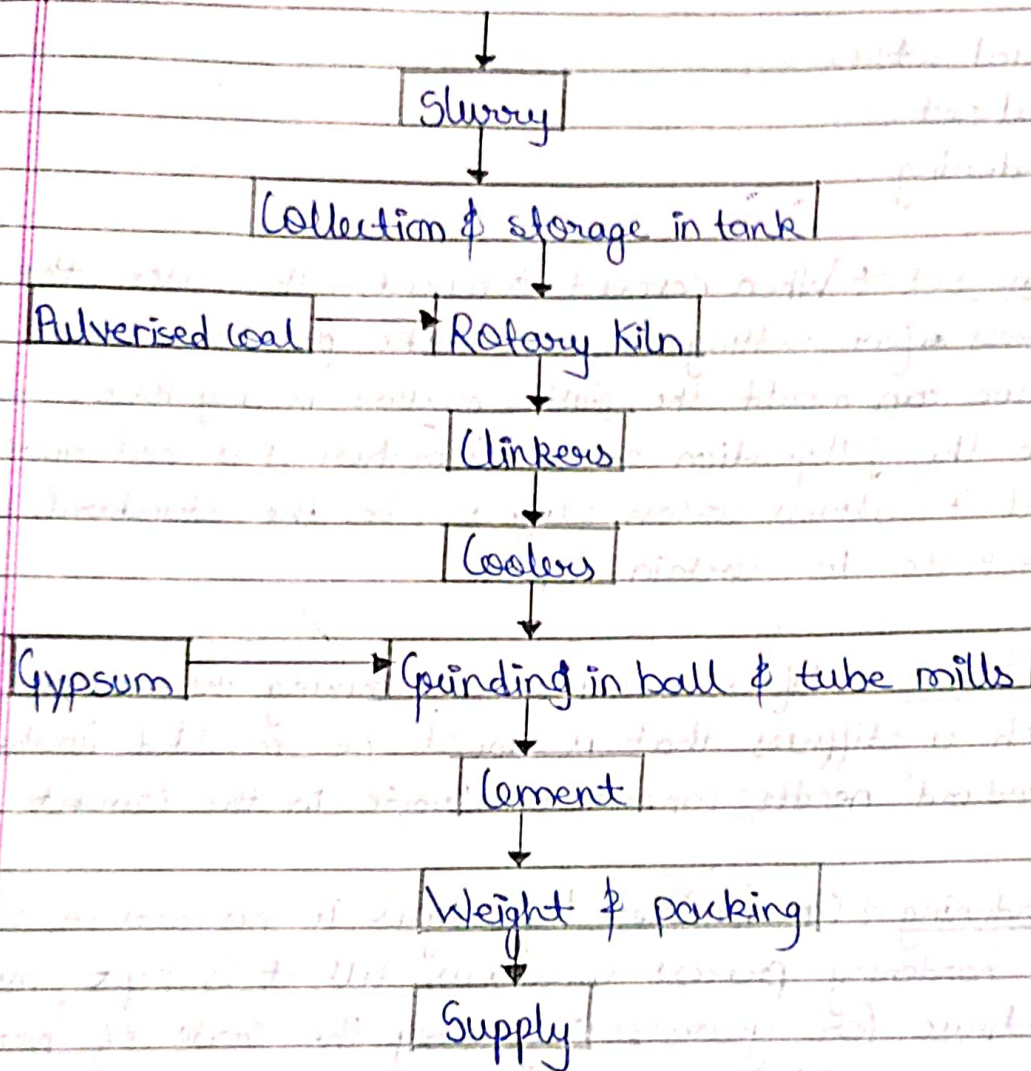
- & Flow diagram of cement manufacturing process





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* —> Setting & Hardening of Portland Cement

- Stiffening of original plastic mass when cement is mixed with water is known as setting & developing strength in it is known as hardening. Setting is due to formation of gel which adheres to stones, sand etc. whereas hardening is due to formation of crystalline products on hydration.
- The whole process continues for about 20 days & various compounds like C_3A , C_3S , C_4AF , C_2S etc. are formed. The hardening process may be continued for years if hydration is continued.
- The setting & hardening may occur in 3 stages.

1. Initial set

2. Final set

3. Hardening

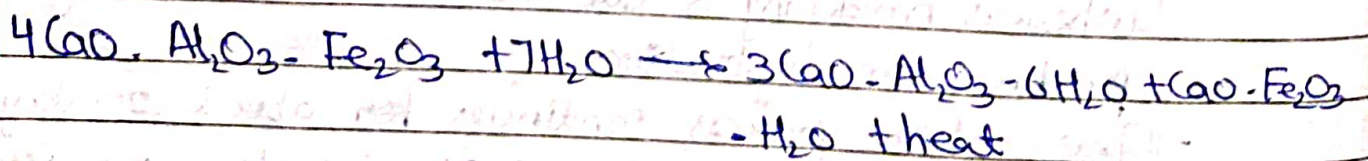
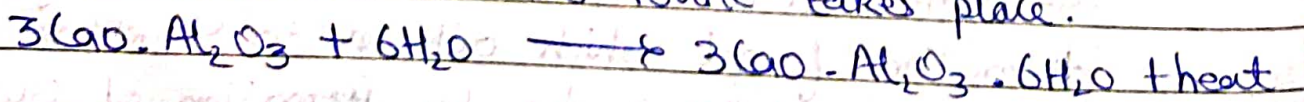
① Initial Set $\Rightarrow$ When cement is mixed with water, the hydrolysis process for setting do not take place for about an hour. So we can mould the plastic mixture in any form. But with time the jellification starts. Further it is not disturbed until it attains certain stiffness. i.e. the standard needle may penetrate to certain depth.

② Final Set $\Rightarrow$ After few hours of mixing the mixture attains such a stiffness that it cannot be moulded further. i.e. the standard needle cannot penetrate in the cement at all.

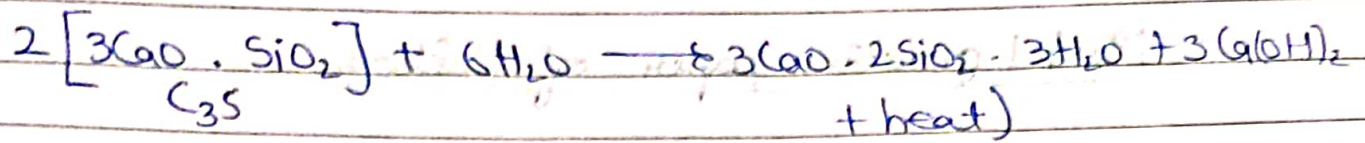
③ Hardening $\Rightarrow$ On further hydrolysis it increases its strength. The hardening process continuous till it is kept moist & may continue for years. Any way the rate of hardening decreases with time.

The reactions involved in the whole process are given below:

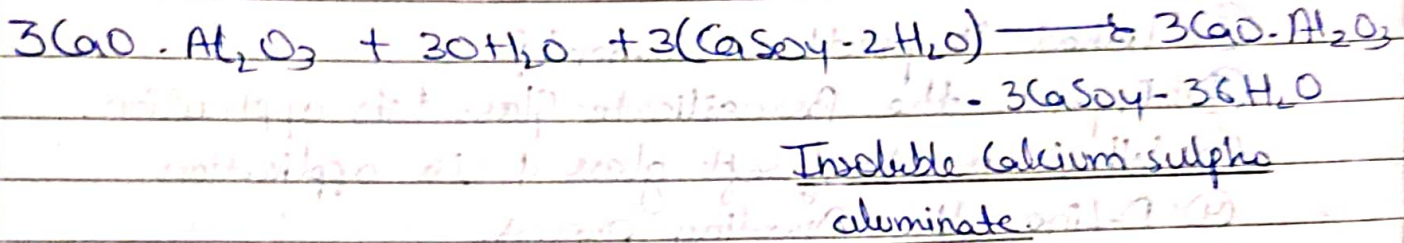
① At first, in one day, hydration of tricalcium aluminate & tetra calcium aluminoferrite takes place.



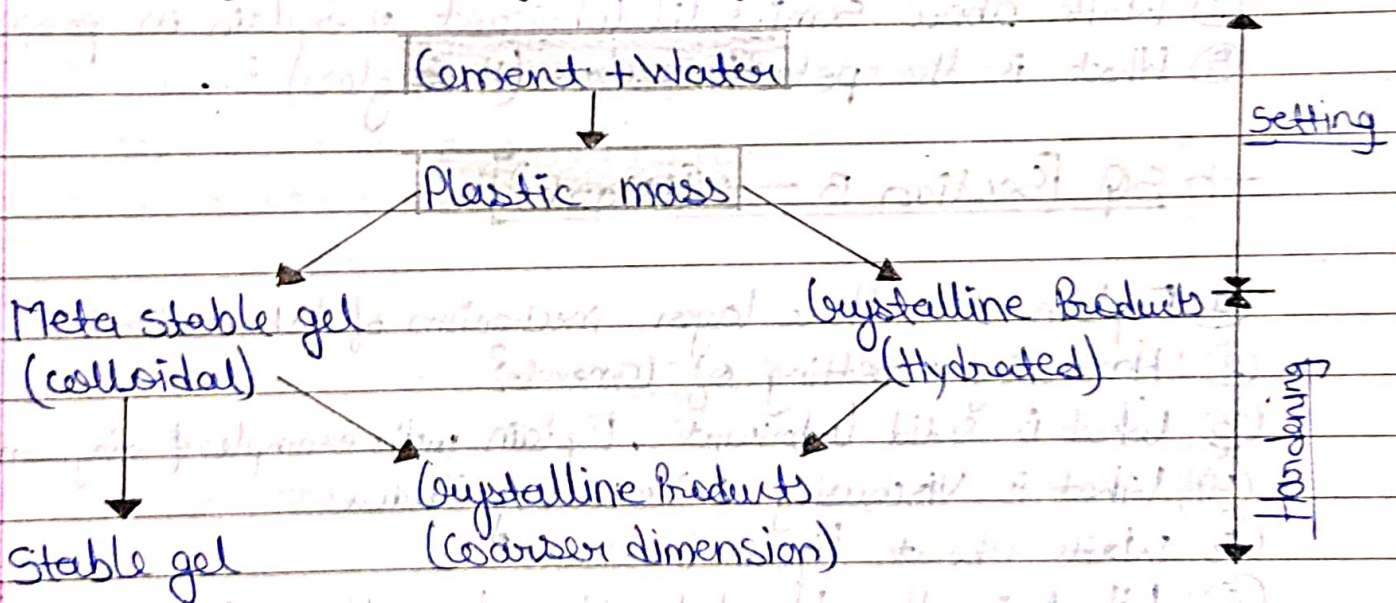
② In 24 hours, the hydration of tricalcium silicate starts & completed in 7 days.



- ③ The aluminates get starts crystalline & dicalcium silicate gets hydrated in 7-28 days. Even hydration of tricalcium silicate is also completed. The addition of gypsum retards the dissolution of C₃A.



- The physical changes during setting & hardening are:-



→ Model question paper of Unit-1

Paper pattern → Section A → 10 Questions (21 Marks)

Section B → 5 or 7 Questions (4 Marks)

Section C → 3 or 5 Questions (10 Marks)

→ VSA [Section A → (21 Marks)]

- ① What is portland cement?
- ② Explain the Borosilicate glass & its application.
- ③ What is the soft glass & its application
- ④ Define the Annealing process.
- ⑤ What is the Lubricant?
- ⑥ Write about pure silica glass? = SiO_2 ^{only made up of.}
- ⑦ Write about Semi-solid lubricant & explain its properties
- ⑧ What is the potashlime glass (Hard glass)? ...

→ SQ [Section B → (4 Marks)]

- ① Explain the three layer mechanism of lubrication?
- ② Hardening & setting of cement?
- ③ What is solid lubricant. Explain with example & diagram.
- ④ What is Viscosity & Viscosity Index?
- ⑤ Write about liquid lubricant?
- ⑥ What is the blended oil. why additives is added in this oil?

→ LQ [Section C → (10 Marks)]

- ① Explain the manufacturing process of cement with diagram & flowchart.
- ② How to manufacture glass with tank furnace process? with Diagram.
- ③ Write a short Note on :- (i) flash & fire point (ii) Reduced Viscosity
(iii) Cloud & Pour Point